

Qualification Pack



Designer Mechanical

QP Code: CSC/Q0405

Version: 1.0

NSQF Level: 5

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CSC/Q0405: Designer Mechanical

Brief Job Description

It involves understanding the customers requirement with respect to the mechanical engineering equipment and establish a design brief, further allocate responsibilities and resources to each activity, and ensure that the complete designing process is completed within agreed deadlines and complying with all relevant regulations, identifying design options, evaluation of design options and their presentation in suitable formats, creating detailed design and models using 2D and 3D softwares for design and obtaining design validations from production and maintenance considerations.

Personal Attributes

Basic communication, numerical and computational abilities. Openness to learning, ability to plan and organize own work and identify and solve problems in the course of working. Understanding the need to take initiative and manage self and work to improve efficiency and effectiveness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CSC/N0402: Make or modify 2D mechanical engineering drawings using CAD system](#)
2. [CSC/N0405: Identify customers requirement and create an engineering design brief](#)
3. [CSC/N0406: Develop plan for engineering design process](#)
4. [CSC/N0407: Create and evaluate mechanical engineering design options](#)
5. [CSC/N0408: Make or modify 3D mechanical engineering models using CAD system](#)
6. [CSC/N1335: Use basic health and safety practices at the workplace](#)
7. [CSC/N1336: Work effectively with others](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Machine Tools
Occupation	Design
Country	India

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NSQF Level	5
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL
Minimum Educational Qualification & Experience	Diploma (Mechanical Engineering) with 1-2 Years of experience Experience in Mechanical Design
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	24/11/2017
Next Review Date	10/08/2025
Deactivation Date	17/01/2025
NSQC Approval Date	18/06/2015
Version	1.0
Reference code on NQR	2015/CCM/GCSC/00248
NQR Version	1.0

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CSC/N0402: Make or modify 2D mechanical engineering drawings using CAD system

Description

This unit covers the skills and knowledge needed to set up and operate a computer aided drawing (CAD) system to produce detailed drawings for engineering activities, in accordance with approved procedures.

Scope

This unit/task covers the following:

- Prepare for 2D mechanical engineering drawings
- Perform set-up activities
- Make or modify 2D mechanical engineering drawings using CAD system

Elements and Performance Criteria

Prepare for 2D mechanical engineering drawings

To be competent, the user/individual on the job must be able to:

- PC1.** use appropriate sources to obtain the technical information relevant to the drawing to be created (Technical information relevant to the drawing to be created: drawing brief; specifications (overall dimensions, materials, special procedures for manufacturing); drawing change or modification request; regulations; existing drawings/designs, sketches, notes from meetings/discussions; standards reference documents (eg. limits and fits, tapping drill charts, contraction allowances)
- PC2.** identify design features, as appropriate to the drawing being produced (Design features: function, materials, clearance, operating environment, quality, aesthetics, interfaces, physical space; tolerances)
- PC3.** ensure that the data and information received is complete and correct
- PC4.** establish the drawing requirements from the data and information received
- PC5.** report and rectify incorrect and inconsistent information in job specification documents as per organization procedures
- PC6.** access and use the correct drawing software
- PC7.** select drafting equipment appropriate to the drawing method chosen
- PC8.** check that all the equipment is correctly connected and in a safe and usable working condition
- PC9.** power up the equipment and activate the appropriate drawing software

Perform set-up activities

To be competent, the user/individual on the job must be able to:

- PC10.** customize system variables, menus and drawing defaults to produce the drawing to the appropriate scale
- PC11.** develop macros as per approved procedures

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- PC12.** set up and check that all peripheral devices are connected and correctly operating and interface with ERP if required is available (Peripheral devices could be: keyboard, mouse, light pen, digitizer/tablet, scanner, printer, plotter, etc.)
- PC13.** set the drawing datum at a convenient point
- PC14.** set up drawing parameters (eg. layers, line types, color, text styles) to company procedures or to suit the drawing produced

Make or modify 2D mechanical engineering drawings using CAD system

To be competent, the user/individual on the job must be able to:

- PC15.** interpret and produce mechanical drawings, using first angle orthographic projections, isometric/oblique projections, third angle orthographic projections, sectional views
- PC16.** apply drafting principles to produce various types of drawings that are consistent with applicable standards and procedures for use in various engineering activities (Types of drawings: detail drawings, sub-assembly drawings, general arrangement drawings, installation drawings, exploded views Standards and procedures: organizational guidelines and procedures, recognized compliance agency/body standards, directives or codes of practice, CAD software standards/protocols, national and/or International standards or directives, customer standards and requirements, health, safety and environmental requirements Engineering Activities: production activities (such as processing of materials, fabrication, finishing, assembly, joining); installation activities (such as commissioning/decommissioning, site preparation, equipment installation); operational activities (such as movement of materials, workplace layouts, work-flow diagrams), maintenance activities (such as planned preventative maintenance, part/sub-asset exchange)
- PC17.** create a drawing template to the required standards, which includes all necessary detail (eg.) using various drawing tools (Drawing template details: layers of drawings, scale, paper size, color setup, line types, dimension system, title, drawing number, date, text styles Drawing Tools: straight lines, hatching and shading on drawings, adding dimensions and text to drawings, producing layers of drawings, symbols and abbreviations, hidden detail, curved/contour lines, angled lines, circles or ellipses; parts lists, geometrical and dimensional tolerance, insertion of standard components, elevation, plane view, side view, sectional views, detail views)
- PC18.** use appropriate terminologies and techniques to create drawings, in the required formats, that are sufficiently and clearly detailed
- PC19.** use keyboard command and pull down menus available in common CAD systems
- PC20.** use codes and other references that follow the required conventions
- PC21.** draw temporary fasteners and rivets
- PC22.** draw components details and assembly drawings
- PC23.** draw piping layouts, gears and machine foundation or base
- PC24.** draw working drawings of jigs and fixtures
- PC25.** draw detailed drawings of dies, moulds and press tools
- PC26.** dimension and label the drawing as per approved procedures
- PC27.** create detailed views using various scales to meet job requirements
- PC28.** ensure that drawings are checked and approved by the appropriate person
- PC29.** produce hard copies of the finished drawings

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- PC30.** check that the drawing is correctly titled and referenced; sawing is correctly titled and referenced
- PC31.** save the drawing to an appropriate storage medium (eg. hard drive, CD/DVD, external storage device)
- PC32.** create a separate backup copy and place it in safe storage
- PC33.** identify component parts list with part name, description of part, material specification or part number, quantities and other details to prepare bill of materials as per organizational guidelines
- PC34.** deal promptly and effectively with problems within control and seek help and guidance from the relevant people if you have problems that they cannot resolve
- PC35.** ensure that changes are completed as required by organizational procedures
- PC36.** shut down the CAD system to a safe condition on completion of the drawing activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** organizational procedures and information systems for retrieving and storing drawing data
- KU11.** system variables that can be customized
- KU12.** procedures and need for customizing identified system variables
- KU13.** applicable drafting standards/procedures
- KU14.** procedures and need for customizing menus and system defaults
- KU15.** procedures and need for developing macros
- KU16.** appropriate projection for the drawing purpose
- KU17.** reasons for selecting the chosen projection
- KU18.** reasons for including auxiliary views in drawings
- KU19.** procedures for producing component, layout and/or assembly drawings
- KU20.** drawing specifications
- KU21.** common symbols used in drawings
- KU22.** how and where to obtain the relevant sources and methods for obtaining any required technical information relevant to the drawing

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- KU23.** methods and procedures used to minimize the chances of infecting a computer with a virus
- KU24.** procedure to follow in case there are corruptions or virus attacks
- KU25.** practices that make systems vulnerable to corruption and damage
- KU26.** basic set-up and operation of the computer system, and the peripheral devices that are used (eg. light pen, digitizer and tablet, printer or plotter, scanner)
- KU27.** how to access the specific computer drawing software to be used, and the use of software manuals and related documents to aid operation of the relevant drawing system
- KU28.** basic principles of engineering manufacturing operations that are used to produce the drawn item (Basic principles of engineering manufacturing operations: casting and forging; fabrication; machining methods; joining processes; assembly and installation methods; limitations of the equipment/processes; kinematics principles relevant to manufacturing of machinery)
- KU29.** types of drawings that may be produced by the software
- KU30.** selection of standard components
- KU31.** functionality of the component being drawn, and its interrelationship with other components and assemblies
- KU32.** how to set up the viewing screen to show multiple views of the drawing to help with drawing creation
- KU33.** standards and conventions that are used for the drawings
- KU34.** how to set up the drawing template parameters
- KU35.** application and use of drawing tools
- KU36.** how to access, recognize and use a wide range of standard components and symbol libraries from the CAD equipment
- KU37.** need for document control
- KU38.** how to save and store drawings
- KU39.** need to create backup copies, and to file them in a separate and safe location
- KU40.** how to produce hard copies of the drawings, and the advantages and disadvantages of printers and plotters

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** undertake numerical operations, and calculations/ formulae (Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
- GS4.** identify and draw various basic, compound and solid shapes as per dimensions given (Basic shapes: square, rectangle, triangle, circle Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder)

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- GS5.** use appropriate units and number systems to express degree of accuracy (Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity)
- GS6.** interpret and express tolerance in terms of limits on dimensions
- GS7.** calculation of the value of angles in a triangle (Angles in a triangle: right-angled, isosceles, equilateral)
- GS8.** convey and share technical information clearly using appropriate language
- GS9.** check and clarify task-related information
- GS10.** liaise with appropriate authorities using correct protocol
- GS11.** communicate with people in respectful form and manner in line with organizational protocol
- GS12.** plan, prioritize and sequence work operations as per job requirements
- GS13.** organize and analyze information relevant to work
- GS14.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS15.** exercise restraint while expressing dissent and during conflict situations
- GS16.** avoid and manage distractions to be disciplined at work
- GS17.** manage own time for achieving better results
- GS18.** work in a team in order to achieve better results
- GS19.** identify and clarify work roles within a team
- GS20.** communicate and cooperate with others in the team for better results
- GS21.** seek assistance from fellow team members
- GS22.** identify problems with work planning, procedures, output and behavior and their implications
- GS23.** prioritize and plan for problem solving
- GS24.** communicate problems appropriately to others
- GS25.** identify sources of information and support for problem solving
- GS26.** seek assistance and support from other sources to solve problems
- GS27.** identify effective resolution techniques
- GS28.** select and apply resolution techniques
- GS29.** seek evidence for problem resolution
- GS30.** undertake and express new ideas and initiatives to others
- GS31.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS32.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS33.** enhance ones competencies in new and different situations and contexts to achieve more
- GS34.** participate in on-the-job and other learning, training and development interventions and assessments
- GS35.** clarify task related information with appropriate personnel or technical adviser
- GS36.** seek to improve and modify own work practices
- GS37.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for 2D mechanical engineering drawings</i>	6	15	-	-
PC1. use appropriate sources to obtain the technical information relevant to the drawing to be created (Technical information relevant to the drawing to be created: drawing brief; specifications (overall dimensions, materials, special procedures for manufacturing); drawing change or modification request; regulations; existing drawings/designs, sketches, notes from meetings/discussions; standards reference documents (eg. limits and fits, tapping drill charts, contraction allowances)	-	2	-	-
PC2. identify design features, as appropriate to the drawing being produced (Design features: function, materials, clearance, operating environment, quality, aesthetics, interfaces, physical space; tolerances)	2	2	-	-
PC3. ensure that the data and information received is complete and correct	-	2	-	-
PC4. establish the drawing requirements from the data and information received	1	2	-	-
PC5. report and rectify incorrect and inconsistent information in job specification documents as per organization procedures	1	2	-	-
PC6. access and use the correct drawing software	1	1	-	-
PC7. select drafting equipment appropriate to the drawing method chosen	1	2	-	-
PC8. check that all the equipment is correctly connected and in a safe and usable working condition	-	1	-	-
PC9. power up the equipment and activate the appropriate drawing software	-	1	-	-
<i>Perform set-up activities</i>	4	10	-	-
PC10. customize system variables, menus and drawing defaults to produce the drawing to the appropriate scale	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. develop macros as per approved procedures	2	2	-	-
PC12. set up and check that all peripheral devices are connected and correctly operating and interface with ERP if required is available (Peripheral devices could be: keyboard, mouse, light pen, digitizer/tablet, scanner, printer, plotter, etc.)	-	2	-	-
PC13. set the drawing datum at a convenient point	-	2	-	-
PC14. set up drawing parameters (eg. layers, line types, color, text styles) to company procedures or to suit the drawing produced	1	2	-	-
<i>Make or modify 2D mechanical engineering drawings using CAD system</i>	20	45	-	-
PC15. interpret and produce mechanical drawings, using first angle orthographic projections, isometric/oblique projections, third angle orthographic projections, sectional views	2	3	-	-
PC16. apply drafting principles to produce various types of drawings that are consistent with applicable standards and procedures for use in various engineering activities (Types of drawings: detail drawings, sub-assembly drawings, general arrangement drawings, installation drawings, exploded views Standards and procedures: organizational guidelines and procedures, recognized compliance agency/body standards, directives or codes of practice, CAD software standards/protocols, national and/or International standards or directives, customer standards and requirements, health, safety and environmental requirements Engineering Activities: production activities (such as processing of materials, fabrication, finishing, assembly, joining); installation activities (such as commissioning/decommissioning, site preparation, equipment installation); operational activities (such as movement of materials, workplace layouts, work-flow diagrams), maintenance activities (such as planned preventative maintenance, part/sub-asset exchange)	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. create a drawing template to the required standards, which includes all necessary detail (eg.) using various drawing tools (Drawing template details: layers of drawings, scale, paper size, color setup, line types, dimension system, title, drawing number, date, text styles Drawing Tools: straight lines, hatching and shading on drawings, adding dimensions and text to drawings, producing layers of drawings, symbols and abbreviations, hidden detail, curved/contour lines, angled lines, circles or ellipses; parts lists, geometrical and dimensional tolerance, insertion of standard components, elevation, plane view, side view, sectional views, detail views)	2	3	-	-
PC18. use appropriate terminologies and techniques to create drawings, in the required formats, that are sufficiently and clearly detailed	2	2	-	-
PC19. use keyboard command and pull down menus available in common CAD systems	1	1	-	-
PC20. use codes and other references that follow the required conventions	1	2	-	-
PC21. draw temporary fasteners and rivets	1	2	-	-
PC22. draw components details and assembly drawings	1	3	-	-
PC23. draw piping layouts, gears and machine foundation or base	1	3	-	-
PC24. draw working drawings of jigs and fixtures	1	3	-	-
PC25. draw detailed drawings of dies, moulds and press tools	1	3	-	-
PC26. dimension and label the drawing as per approved procedures	1	3	-	-
PC27. create detailed views using various scales to meet job requirements	1	2	-	-
PC28. ensure that drawings are checked and approved by the appropriate person	-	1	-	-
PC29. produce hard copies of the finished drawings	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. check that the drawing is correctly titled and referenced; sawing is correctly titled and referenced	-	2	-	-
PC31. save the drawing to an appropriate storage medium (eg. hard drive, CD/DVD, external storage device)	-	1	-	-
PC32. create a separate backup copy and place it in safe storage	-	1	-	-
PC33. identify component parts list with part name, description of part, material specification or part number, quantities and other details to prepare bill of materials as per organizational guidelines	2	2	-	-
PC34. deal promptly and effectively with problems within control and seek help and guidance from the relevant people if you have problems that they cannot resolve	-	2	-	-
PC35. ensure that changes are completed as required by organizational procedures	1	1	-	-
PC36. shut down the CAD system to a safe condition on completion of the drawing activities	-	1	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0402
NOS Name	Make or modify 2D mechanical engineering drawings using CAD system
Sector	Capital Goods
Sub-Sector	Machine Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Design
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/11/2017
Next Review Date	31/03/2022
NSQC Clearance Date	19/05/2015

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CSC/N0405: Identify customers requirement and create an engineering design brief

Description

This unit is about identifying the engineering design requirements of the customer and creating an engineering design brief, as per approved procedures and using cost optimization techniques.

Scope

This unit/task covers the following:

- Work safely
- Identify and interpret engineering design requirements of customer
- Identify, interpret and communicate manufacturing machinery design requirements of customer

Elements and Performance Criteria

Work safely

To be competent, the user/individual on the job must be able to:

- PC1.** work safely at all times, complying with health and safety, environmental and other relevant regulations and guidelines
- PC2.** check that all safety mechanisms are in place and that the equipment is set correctly for the required operations
- PC3.** adhere to procedures or systems in place for health and safety, including personal protective equipment and other relevant safety regulations and procedures to contribute to a safe work environment
- PC4.** wear the appropriate protective clothing and equipment, and keep the work area clean and tidy
- PC5.** follow safe practice/approved setting up procedures at all times

Identify and interpret engineering design requirements of customer

To be competent, the user/individual on the job must be able to:

- PC6.** gather accurate information on the requirements of the customer from various sources (Sources: purchase order, quotation documents submitted to customer, customer interaction(self or others), sales representative/application engineer/proposal engineer, existing designs, research, suppliers, process and manufacturing technologies)
- PC7.** confirm the customer's objectives for the engineering products or processes
- PC8.** identify any unique or specific features that need particular consideration
- PC9.** determine the feasibility of achieving the customer's requirements
- PC10.** confirm the requirements and other relevant issues with the customer
- PC11.** record all relevant information in the appropriate information systems for future use

Identify, interpret and communicate manufacturing machinery design requirements of customer

To be competent, the user/individual on the job must be able to:

- PC12.** confirm the operational and functional requirements and quality criteria of the design

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- PC13.** obtain clarification from relevant people regarding any aspect of the requirement that is not clear
- PC14.** identify clearly any design constraints (Design brief constraints: customer acceptability, departmental constraints, available technologies, environmental/sustainability, delivery schedule, legal, logistical, financial, international/national standards or directives, safety, capacity, capability, copyright, commercial/branding, ease of maintenance)
- PC15.** create the design brief in a draft form and discuss any changes required with the relevant people (Design brief details: confirmation of objectives, draft design concepts, supporting calculations and data, overall functionality, feasibility of achieving requirements, any special features, detail of specific issues for consideration (such as product safety, health and safety, impending regulation changes, emerging technologies), design process, product life cycle requirements, support required)
- PC16.** ensure that the design brief captures all the requirements of the customer
- PC17.** ensure that the design brief and specification meets relevant regulations, directives and guidelines (Regulations, directives and guidelines: organizational guidelines and codes of practice; recognized compliance agency/body's standards; equipment manufacturer's operating specification/range; customer standards and requirements; national or International standards or directives; health, safety and environmental requirements)
- PC18.** save the design brief and communicate it to the relevant people, as per organizational process (Communicate via any of the following: a verbal report, electronic mail, presentation, computer generated report, specific company document)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** company systems for recording design information
- KU11.** importance of using the company information systems
- KU12.** limits of learners own authority, and to whom should they report if they have problems that they cannot resolve
- KU13.** purpose of a design brief and its importance
- KU14.** how to obtain details of the specification of the product or process to be designed
- KU15.** various sources for information for the design brief
- KU16.** when can a customer be consulted on a design brief

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- KU17.** how to obtain and interpret legislative and regulatory documentation
- KU18.** types of design features that should be considered unique or specific
- KU19.** factors that affect the feasibility of achieving a customer's requirements
- KU20.** how to assess the feasibility of achieving the customer's requirements
- KU21.** information and level of detail to be included in a design brief
- KU22.** how to prepare a brief confirming the requirements of the customer
- KU23.** importance of identifying design constraints
- KU24.** different types of design briefs
- KU25.** who should be informed and consulted on the various aspects of a design brief and specification
- KU26.** regulations, directives and guidelines that are relevant
- KU27.** how to obtain information on regulations, directives and guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** undertake numerical operations, geometry and calculations/ formulae (Arithmetic: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
- GS4.** use appropriate measuring techniques
- GS5.** express numerical solutions to a degree of accuracy that is appropriate to the value being calculated (Degree of accuracy: correct to three significant figures, correct to two decimal places, express a decimal fraction in standard form, express tolerance in terms of limits of size)
- GS6.** use a calculator to raise a number to a power and determine square roots
- GS7.** use formulae to complete transpositions and solve problems (Transpositions: involving addition, subtraction, multiplication and division in any combination using a maximum of three terms, for example Ohms Law, substitution of known values)
- GS8.** use algebraic expressions to solve linear equations
- GS9.** plot and interpret straight line graphs
- GS10.** convey and share technical information clearly using appropriate language
- GS11.** check and clarify task-related information
- GS12.** liaise with appropriate authorities using correct protocol
- GS13.** communicate with people in respectful form and manner in line with organizational protocol
- GS14.** listen to questions and concerns of the customer and provide resolution in a respectful manner as per organizational guidelines
- GS15.** use basic office applications like spread sheet, word processor, presentations
- GS16.** use ERP software and other organizational software specific to quality function

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- GS17.** use email to communicate within the organization as per organization guidelines
- GS18.** be well dressed and groomed
- GS19.** put forward ones point of view in a convincing manner
- GS20.** plan, prioritize and sequence work operations as per job requirements
- GS21.** organize and analyze information relevant to work
- GS22.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS23.** communicate with customers following organizational protocols and practices generating customer satisfaction and delight
- GS24.** undertake clear and open communication with customers for trust building and clarifying and managing expectations of customers
- GS25.** respond to customer expectation promptly and recognizing and communicating limits of ones authority
- GS26.** deal with customer feedback
- GS27.** handle customer disgruntlement and dissatisfaction
- GS28.** work taking responsibility for own work outcomes
- GS29.** adhere to work timings, dress code and other organizational policies
- GS30.** work following laid down rules, procedures, instructions and policies
- GS31.** conduct oneself express dissent during conflict situations while exercising restraint
- GS32.** avoid and manage distractions to be disciplined at work
- GS33.** work by time management for achieving better results
- GS34.** work in a team in order to achieve better results
- GS35.** identify and clarify work roles within a team
- GS36.** communicate and cooperate with others in the team
- GS37.** seek assistance from fellow team members
- GS38.** co-ordinate across teams and personnel for getting work done
- GS39.** identify problems with work planning, procedures, output and behavior and their implications
- GS40.** prioritize and plan for problem solving
- GS41.** communicate problems appropriately to others
- GS42.** identify sources of information and support for problem solving
- GS43.** seek assistance and support from other sources to solve problems
- GS44.** identify effective resolution techniques
- GS45.** select and apply resolution techniques
- GS46.** seek evidence for problem resolution
- GS47.** inspect quality of own or other employee's work
- GS48.** analyze information according to enterprise and work requirements
- GS49.** use diagnostic skills to identify and determine causes of faults, including interpretation of in-built fault indicators and error codes
- GS50.** take decisions within if within own jurisdiction or take approval for case outside own jurisdiction

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- GS51.** work towards achieving better results for self, others and organization by displaying initiative and enterprise
- GS52.** undertake and express new ideas and initiatives to others
- GS53.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS54.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS55.** achieve more by applying ones competencies in new and different situations and contexts to achieve more
- GS56.** identify potential business opportunities for the company
- GS57.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS58.** participate in on-the-job and other learning, training and development interventions and assessments
- GS59.** clarify task related information with appropriate personnel or technical adviser
- GS60.** seek to improve and modify own work practices

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work safely</i>	10	18	-	-
PC1. work safely at all times, complying with health and safety, environmental and other relevant regulations and guidelines	2	4	-	-
PC2. check that all safety mechanisms are in place and that the equipment is set correctly for the required operations	2	3	-	-
PC3. adhere to procedures or systems in place for health and safety, including personal protective equipment and other relevant safety regulations and procedures to contribute to a safe work environment	2	4	-	-
PC4. wear the appropriate protective clothing and equipment, and keep the work area clean and tidy	2	4	-	-
PC5. follow safe practice/approved setting up procedures at all times	2	3	-	-
<i>Identify and interpret engineering design requirements of customer</i>	12	22	-	-
PC6. gather accurate information on the requirements of the customer from various sources (Sources: purchase order, quotation documents submitted to customer, customer interaction(self or others), sales representative/application engineer/proposal engineer, existing designs, research, suppliers, process and manufacturing technologies)	2	3	-	-
PC7. confirm the customer's objectives for the engineering products or processes	2	4	-	-
PC8. identify any unique or specific features that need particular consideration	2	4	-	-
PC9. determine the feasibility of achieving the customer's requirements	2	4	-	-
PC10. confirm the requirements and other relevant issues with the customer	2	4	-	-
PC11. record all relevant information in the appropriate information systems for future use	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify, interpret and communicate manufacturing machinery design requirements of customer</i>	14	24	-	-
PC12. confirm the operational and functional requirements and quality criteria of the design	2	4	-	-
PC13. obtain clarification from relevant people regarding any aspect of the requirement that is not clear	2	3	-	-
PC14. identify clearly any design constraints (Design brief constraints: customer acceptability, departmental constraints, available technologies, environmental/sustainability, delivery schedule, legal, logistical, financial, international/national standards or directives, safety, capacity, capability, copyright, commercial/branding, ease of maintenance)	2	4	-	-
PC15. create the design brief in a draft form and discuss any changes required with the relevant people (Design brief details: confirmation of objectives, draft design concepts, supporting calculations and data, overall functionality, feasibility of achieving requirements, any special features, detail of specific issues for consideration (such as product safety, health and safety, impending regulation changes, emerging technologies), design process, product life cycle requirements, support required)	2	4	-	-
PC16. ensure that the design brief captures all the requirements of the customer	2	3	-	-
PC17. ensure that the design brief and specification meets relevant regulations, directives and guidelines (Regulations, directives and guidelines: organizational guidelines and codes of practice; recognized compliance agency/body's standards; equipment manufacturer's operating specification/range; customer standards and requirements; national or International standards or directives; health, safety and environmental requirements)	2	3	-	-
PC18. save the design brief and communicate it to the relevant people, as per organizational process (Communicate via any of the following: a verbal report, electronic mail, presentation, computer generated report, specific company document)	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	36	64	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0405
NOS Name	Identify customers requirement and create an engineering design brief
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Design
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	24/11/2017
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Qualification Pack

CSC/N0406: Develop plan for engineering design process

Description

This unit covers planning for engineering design process as per approved processes, for a smooth and timely delivery of the final design. It covers the identification of the design activities that needed to be undertaken, to allocate responsibilities and resources to each activity.

Scope

This unit/task covers the following:

- Work safely
- Plan for the engineering design process

Elements and Performance Criteria

Work safely

To be competent, the user/individual on the job must be able to:

- PC1.** work safely at all times, complying with health and safety, environmental and other relevant regulations and guidelines
- PC2.** check that all safety mechanisms are in place and that the equipment is set correctly for the required operations
- PC3.** adhere to procedures or systems in place for health and safety, including personal protective equipment and other relevant safety regulations and procedures to contribute to a safe work environment
- PC4.** wear the appropriate protective clothing and equipment, and keep the work area clean and tidy
- PC5.** follow safe practice/approved setting up procedures at all times

Plan for the engineering design process

To be competent, the user/individual on the job must be able to:

- PC6.** identify the design activities to be undertaken (Design activities: e.g. confirmation of requirements; production and review of detailed design/s; review of reference materials; development of models (such as software, physical); production and review of high level design/s; obtaining final approval, etc.)
- PC7.** identify the activities that make up the design process (Activities that make up the design process: disseminating information, change management, obtaining resources, configuration management, reviewing design/s, resource procurement)
- PC8.** establish the responsibilities for developing specific aspects of the design process
- PC9.** identify the activities that make up the design process (Activities that make up the design process: disseminating information, change management, obtaining resources, configuration management, reviewing design/s, resource procurement)
- PC10.** communicate information to the appropriate people using various company specific media (Media: verbal report, electronic mail, presentation, computer generated report, specific company document)

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- PC11.** identify the resources necessary to undertake the design process agree procedures for disseminating information on the designs
- PC12.** identify any potentially critical problems and include contingency plans for the same
- PC13.** develop a schedule for the design process
- PC14.** agree the schedule with the appropriate people
- PC15.** establish priorities for completion of the design process within deadlines
- PC16.** ensure that the design process complies with all relevant regulations, directives and guidelines Regulations, directives and guidelines: organizational guidelines and procedures; recognized compliance agency/body's standards, directives or codes of practice; equipment manufacturer's operating specification/manual; customers requirements; international and or national standards; health, safety and environmental requirements
- PC17.** obtain approvals of the relevant people for the design plan
- PC18.** establish version control for the document
- PC19.** save and store the design documentation as per organizational guidelines
- PC20.** communicate information to the appropriate people using various company specific media (Media: verbal report, electronic mail, presentation, computer generated report, specific company document)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** the organizational activities required for the design process
- KU11.** importance of establishing and recording responsibilities
- KU12.** who should have responsibility for developing different parts of a design
- KU13.** various procedures that can be used in the design process
- KU14.** factors that should be taken into account for disseminating information
- KU15.** types of problem that could occur during the design process
- KU16.** why it is important to have contingency plans
- KU17.** what should be included in contingency plans
- KU18.** how to priorities and schedule design activities
- KU19.** how to obtain information on resources

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- KU20.** how to determine what resources are necessary
- KU21.** how to determine the availability of resources
- KU22.** organizational and regulatory, directives and guidelines that are relevant
- KU23.** how to obtain information on relevant regulations, directives and guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** undertake numerical operations, geometry and calculations/formulae (Arithmetic: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
- GS4.** use appropriate measuring techniques
- GS5.** express numerical solutions to a degree of accuracy that is appropriate to the value being calculated (Degree of accuracy: correct to three significant figures, correct to two decimal places, express a decimal fraction in standard form, express tolerance in terms of limits of size)
- GS6.** use a calculator to raise a number to a power and determine square roots
- GS7.** use formulae to complete transpositions and solve problems (Transpositions: involving addition, subtraction, multiplication and division in any combination using a maximum of three terms, for example Ohms Law, substitution of known values)
- GS8.** use algebraic expressions to solve linear equations
- GS9.** plot and interpret straight line graphs
- GS10.** write a small program which consists of all the machine functions
- GS11.** convey and share technical information clearly using appropriate language
- GS12.** check and clarify task-related information
- GS13.** liaise with appropriate authorities using correct protocol
- GS14.** communicate with people in respectful form and manner in line with organizational protocol
- GS15.** listen to questions and concerns of the customer and provide resolution in a respectful manner as per organizational guidelines
- GS16.** use basic office applications like spread sheet, word processor, presentations
- GS17.** use ERP software and other organizational software specific to quality function
- GS18.** use email to communicate within the organization as per organization guidelines
- GS19.** be well dressed and groomed
- GS20.** put forward ones point of view in a convincing manner
- GS21.** plan, prioritize and sequence work operations as per job requirements
- GS22.** organize and analyze information relevant to work
- GS23.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time

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- GS24.** communicate with customers following organizational protocols and practices generating customer satisfaction and delight
- GS25.** undertake clear and open communication with customers for trust building and clarifying and managing expectations of customers
- GS26.** respond to customer expectation promptly and recognizing and communicating limits of ones authority
- GS27.** deal with customer feedback
- GS28.** handle customer disgruntlement and dissatisfaction
- GS29.** work taking responsibility for own work outcomes
- GS30.** adhere to work timings, dress code and other organizational policies
- GS31.** work following laid down rules, procedures, instructions and policies
- GS32.** conduct oneself express dissent during conflict situations while exercising restraint
- GS33.** avoid and manage distractions to be disciplined at work
- GS34.** work by time management for achieving better results
- GS35.** work in a team in order to achieve better results
- GS36.** identify and clarify work roles within a team
- GS37.** communicate and cooperate with others in the team
- GS38.** seek assistance from fellow team members
- GS39.** co-ordinate across teams and personnel for getting work done
- GS40.** identify problems with work planning, procedures, output and behavior and their implications
- GS41.** prioritize and plan for problem solving
- GS42.** communicate problems appropriately to others
- GS43.** identify sources of information and support for problem solving
- GS44.** seek assistance and support from other sources to solve problems
- GS45.** identify effective resolution techniques
- GS46.** select and apply resolution techniques
- GS47.** seek evidence for problem resolution
- GS48.** inspect quality of own or other employee's work
- GS49.** analyze information according to enterprise and work requirements
- GS50.** use diagnostic skills to identify and determine causes of faults, including interpretation of in-built fault indicators and error codes
- GS51.** take decisions within if within own jurisdiction or take approval for case outside own jurisdiction
- GS52.** work towards achieving better results for self, others and organization by displaying initiative and enterprise
- GS53.** undertake and express new ideas and initiatives to others
- GS54.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS55.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS56.** achieve more by applying ones competencies in new and different situations and contexts to achieve more

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- GS57.** identify potential business opportunities for the company
- GS58.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS59.** participate in on-the-job and other learning, training and development interventions and assessments
- GS60.** clarify task related information with appropriate personnel or technical adviser
- GS61.** seek to improve and modify own work practices

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work safely</i>	9	14	-	-
PC1. work safely at all times, complying with health and safety, environmental and other relevant regulations and guidelines	2	3	-	-
PC2. check that all safety mechanisms are in place and that the equipment is set correctly for the required operations	2	2	-	-
PC3. adhere to procedures or systems in place for health and safety, including personal protective equipment and other relevant safety regulations and procedures to contribute to a safe work environment	2	3	-	-
PC4. wear the appropriate protective clothing and equipment, and keep the work area clean and tidy	2	3	-	-
PC5. follow safe practice/approved setting up procedures at all times	1	3	-	-
<i>Plan for the engineering design process</i>	29	48	-	-
PC6. identify the design activities to be undertaken (Design activities: e.g. confirmation of requirements; production and review of detailed design/s; review of reference materials; development of models (such as software, physical); production and review of high level design/s; obtaining final approval, etc.)	2	3	-	-
PC7. identify the activities that make up the design process (Activities that make up the design process: disseminating information, change management, obtaining resources, configuration management, reviewing design/s, resource procurement)	2	3	-	-
PC8. establish the responsibilities for developing specific aspects of the design process	2	4	-	-
PC9. identify the activities that make up the design process (Activities that make up the design process: disseminating information, change management, obtaining resources, configuration management, reviewing design/s, resource procurement)	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. communicate information to the appropriate people using various company specific media (Media: verbal report, electronic mail, presentation, computer generated report, specific company document)	2	4	-	-
PC11. identify the resources necessary to undertake the design process agree procedures for disseminating information on the designs	2	3	-	-
PC12. identify any potentially critical problems and include contingency plans for the same	2	3	-	-
PC13. develop a schedule for the design process	2	3	-	-
PC14. agree the schedule with the appropriate people	2	4	-	-
PC15. establish priorities for completion of the design process within deadlines	2	3	-	-
PC16. ensure that the design process complies with all relevant regulations, directives and guidelines Regulations, directives and guidelines: organizational guidelines and procedures; recognized compliance agency/body's standards, directives or codes of practice; equipment manufacturer's operating specification/manual; customers requirements; international and or national standards; health, safety and environmental requirements	2	3	-	-
PC17. obtain approvals of the relevant people for the design plan	2	4	-	-
PC18. establish version control for the document	2	3	-	-
PC19. save and store the design documentation as per organizational guidelines	2	2	-	-
PC20. communicate information to the appropriate people using various company specific media (Media: verbal report, electronic mail, presentation, computer generated report, specific company document)	1	3	-	-
NOS Total	38	62	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0406
NOS Name	Develop plan for engineering design process
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Design
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	24/11/2017
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Qualification Pack

CSC/N0407: Create and evaluate mechanical engineering design options

Description

It involves understanding the customers requirement with respect to the mechanical engineering equipment and establish a design brief, further allocate responsibilities and resources to each activity, and ensure that the complete designing process is completed within agreed deadlines and complying with all relevant regulations, identifying design options, evaluation of design options and their presentation in suitable formats, creating detailed design and models using 2D and 3D softwares for design and obtaining design validations from production and maintenance considerations.

Scope

This unit/task covers the following:

- Create and presenting engineering design options
- Evaluate engineering design options

Elements and Performance Criteria

Create and presenting engineering design options

To be competent, the user/individual on the job must be able to:

- PC1.** obtain and review existing information with reference to the specified design requirements (Existing information: drawing brief; modification request; regulations; calculations, previous drawings/designs, sketches, previous test/trial data, modelling data, standards reference documents, notes from meetings/discussions) (Design requirements as per the following: customer requirements; legal/copyright considerations; design brief; regulatory requirements; design constraints)
- PC2.** prepare outline ideas for the designs
- PC3.** obtain agreement from relevant people
- PC4.** carry out the design process, utilizing the appropriate technology
- PC5.** document all facets of the design activity
- PC6.** communicate the outcomes of the design process to the appropriate people via various media used in the organisation (Media: a verbal report; presentation; computer generated report; specific company document)
- PC7.** deliver the designs in the appropriate format
- PC8.** ensure that the design cannot be changed or amended without authorization
- PC9.** confirm and agree understanding of the design requirements
- PC10.** deal with problems relating to the design requirements and agreed solutions
- PC11.** identify design options which will meet requirements and the design Specification
- PC12.** create designs that meet the customer's requirements as specified in the design brief for the engineering product or process

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- PC13.** apply approved general and sub-sector specific engineering concepts, processes, principles to achieve the design brief (Engineering or manufacturing principles and concepts: metals, plastic, ceramics materials and their properties; basic metallurgy and heat treatment; thermal properties; thermal stress analysis-heat treatment diagram/process; structural engineering/analytics; finite element analysis; manufacturing technologies; welding principles; fabrication principles; kinematics and dynamics principles; design calculations like pressure, force, capacity etc.; trigonometry, geometry, dimensional and geometric tolerance; general engineering drawing)
- PC14.** apply the principles of dynamics and kinematics to ensure that design options will work
- PC15.** ensure that the design options are practical
- PC16.** prepare costings and timescale and ensure they are acceptable
- PC17.** obtain suitable advice and guidance to assist in the design work
- PC18.** present the designs in suitable formats and with sufficient information to allow the customer to assess them
- PC19.** ensure that the designs comply with all relevant regulations, standards directives or codes of practice (Regulations, standards directives or codes of practice: organisational guidelines and procedures; recognised compliance agency/body's standards, directives or codes of practice; equipment manufacturer's operating specification/range; customer standards and requirements; national and/or International standards or directives; health, safety and environmental Requirements)
- PC20.** deal promptly and effectively with problems within your control and seek help and guidance from the relevant people if you have problems that you cannot resolve
- PC21.** ensure that the designs are protected in line with organizational procedures

Evaluate engineering design options

To be competent, the user/individual on the job must be able to:

- PC22.** obtain clear criteria on which to base the evaluation (Criteria for evaluating designs: function; financial constraints; manufacturing or installation requirements; installation or commissioning requirements; building redundancy into the design; appropriate materials; technology; aesthetics; performance/capability; reliability; life cycle of product, system or process; compatibility; maintenance and repair; product features; availability of resources; characteristics; corporate branding; components to be used; any interface requirements; future customer support; timescales; diversity/alternatives; safety; environmental/sustainability factors)
- PC23.** obtain the necessary information from the available sources
- PC24.** evaluate the design against the established criteria, using appropriate evaluation methods (Evaluation methods: market research; software simulation; analysis of the design documentation; simulation; model; prototype assessment; pilot trial; small-scale production)
- PC25.** make recommendations on various design options, and communicate the results of the evaluation to the relevant people

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

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- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU6.** relevant people and their responsibilities within the work area
- KU7.** escalation matrix and procedures for reporting work and employment related issues
- KU8.** documentation and related procedures applicable in the context of employment and work
- KU9.** importance and purpose of documentation in context of employment and work
- KU10.** the organizational activities required for the design process
- KU11.** organizational procedures and information systems for storing design data and configuration management
- KU12.** national and international standards and conventions that are used for the design
- KU13.** underlying general and sub-sector specific engineering or manufacturing principles and concepts required to produce fit for purpose designs (Engineering or manufacturing principles and concepts: metals, plastic, ceramics materials and their properties; basic metallurgy and heat treatment; thermal properties; thermal stress analysis-heat treatment diagram/process; structural engineering/analytics; finite element analysis; manufacturing technologies; welding principles; fabrication principles; kinematics and dynamics principles; design calculations like pressure, force, capacity etc.; trigonometry, geometry, dimensional and geometric tolerance; general engineering drawing)
- KU14.** functionality of the design including any interrelationships required with other components/products/systems or technologies
- KU15.** working knowledge and understanding of the relative costs likely to be incurred during the development and production of the design
- KU16.** regulations, standards, directives and codes of practice that are relevant, and any implications they have on the design
- KU17.** methods for achieving different types of design
- KU18.** design formats that are most suitable to meet the design team's needs
- KU19.** potential risks to a design, and how can it be protected
- KU20.** importance of establishing and recording responsibilities
- KU21.** how and where to obtain the design brief/specification
- KU22.** procedures used for making changes or amendments to the design
- KU23.** sources of advice and guidance on designs
- KU24.** how to present designs to the customer
- KU25.** need for effective document and data control and the implications if these are not applied
- KU26.** patent, copyright and intellectual property issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, health and safety instructions, memos, etc. applicable to the job in English and/or local language

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- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** undertake numerical operations, geometry and calculations/ formulae (Arithmetic: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
- GS4.** use appropriate measuring techniques
- GS5.** express numerical solutions to a degree of accuracy that is appropriate to the value being calculated (Degree of accuracy: correct to three significant figures, correct to two decimal places, express a decimal fraction in standard form, express tolerance in terms of limits of size)
- GS6.** use a calculator to raise a number to a power and determine square roots
- GS7.** use formulae to complete transpositions and solve problems (Transpositions: involving addition, subtraction, multiplication and division in any combination using a maximum of three terms, for example Ohms Law, substitution of known values)
- GS8.** use algebraic expressions to solve linear equations
- GS9.** plot and interpret straight line graphs
- GS10.** write a small program which consists of all the machine functions
- GS11.** convey and share technical information clearly using appropriate language
- GS12.** check and clarify task-related information
- GS13.** liaise with appropriate authorities using correct protocol
- GS14.** communicate with people in respectful form and manner in line with organizational protocol
- GS15.** listen to questions and concerns of the customer and provide resolution in a respectful manner as per organizational guidelines
- GS16.** use basic office applications like spread sheet, word processor, presentations
- GS17.** use ERP software and other organizational software specific to quality function
- GS18.** use email to communicate within the organization as per organization guidelines
- GS19.** be well dressed and groomed
- GS20.** put forward one's point of view in a convincing manner
- GS21.** plan, prioritize and sequence work operations as per job requirements
- GS22.** organize and analyze information relevant to work
- GS23.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS24.** communicate with customers following organizational protocols and practices generating customer satisfaction and delight
- GS25.** undertake clear and open communication with customers for trust building and clarifying and managing expectations of customers
- GS26.** respond to customer expectation promptly and recognizing and communicating limits of ones authority
- GS27.** deal with customer feedback
- GS28.** handle customer disgruntlement and dissatisfaction
- GS29.** work taking responsibility for own work outcomes
- GS30.** adhere to work timings, dress code and other organizational policies
- GS31.** work following laid down rules, procedures, instructions and policies

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- GS32.** conduct oneself express dissent during conflict situations while exercising restraint
- GS33.** avoid and manage distractions to be disciplined at work
- GS34.** work by time management for achieving better results
- GS35.** work in a team in order to achieve better results
- GS36.** identify and clarify work roles within a team
- GS37.** communicate and cooperate with others in the team
- GS38.** seek assistance from fellow team members
- GS39.** co-ordinate across teams and personnel for getting work done
- GS40.** identify problems with work planning, procedures, output and behavior and their implications
- GS41.** prioritize and plan for problem solving
- GS42.** communicate problems appropriately to others
- GS43.** identify sources of information and support for problem solving
- GS44.** seek assistance and support from other sources to solve problems
- GS45.** identify effective resolution techniques
- GS46.** select and apply resolution techniques
- GS47.** seek evidence for problem resolution
- GS48.** inspect quality of own or other employee's work
- GS49.** analyze information according to enterprise and work requirements
- GS50.** use diagnostic skills to identify and determine causes of faults, including interpretation of in-built fault indicators and error codes
- GS51.** take decisions within if within own jurisdiction or take approval for case outside own jurisdiction
- GS52.** work towards achieving better results for self, others and organization by displaying initiative and enterprise
- GS53.** undertake and express new ideas and initiatives to others
- GS54.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS55.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS56.** achieve more by applying ones competencies in new and different situations and contexts to achieve more
- GS57.** identify potential business opportunities for the company
- GS58.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS59.** participate in on-the-job and other learning, training and development interventions and assessments
- GS60.** clarify task related information with appropriate personnel or technical adviser
- GS61.** seek to improve and modify own work practices

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Create and presenting engineering design options</i>	29	53	-	-
PC1. obtain and review existing information with reference to the specified design requirements (Existing information: drawing brief; modification request; regulations; calculations, previous drawings/designs, sketches, previous test/trial data, modelling data, standards reference documents, notes from meetings/discussions) (Design requirements as per the following: customer requirements; legal/copyright considerations; design brief; regulatory requirements; design constraints)	1	2	-	-
PC2. prepare outline ideas for the designs	1	3	-	-
PC3. obtain agreement from relevant people	2	1	-	-
PC4. carry out the design process, utilizing the appropriate technology	2	3	-	-
PC5. document all facets of the design activity	2	2	-	-
PC6. communicate the outcomes of the design process to the appropriate people via various media used in the organisation (Media: a verbal report; presentation; computer generated report; specific company document)	1	2	-	-
PC7. deliver the designs in the appropriate format	2	3	-	-
PC8. ensure that the design cannot be changed or amended without authorization	1	2	-	-
PC9. confirm and agree understanding of the design requirements	2	2	-	-
PC10. deal with problems relating to the design requirements and agreed solutions	1	2	-	-
PC11. identify design options which will meet requirements and the design Specification	1	3	-	-
PC12. create designs that meet the customer's requirements as specified in the design brief for the engineering product or process	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. apply approved general and sub-sector specific engineering concepts, processes, principles to achieve the design brief (Engineering or manufacturing principles and concepts: metals, plastic, ceramics materials and their properties; basic metallurgy and heat treatment; thermal properties; thermal stress analysis-heat treatment diagram/process; structural engineering/analytics; finite element analysis; manufacturing technologies; welding principles; fabrication principles; kinematics and dynamics principles; design calculations like pressure, force, capacity etc.; trigonometry, geometry, dimensional and geometric tolerance; general engineering drawing)	1	4	-	-
PC14. apply the principles of dynamics and kinematics to ensure that design options will work	1	4	-	-
PC15. ensure that the design options are practical	2	2	-	-
PC16. prepare costings and timescale and ensure they are acceptable	1	3	-	-
PC17. obtain suitable advice and guidance to assist in the design work	2	2	-	-
PC18. present the designs in suitable formats and with sufficient information to allow the customer to assess them	1	3	-	-
PC19. ensure that the designs comply with all relevant regulations, standards directives or codes of practice (Regulations, standards directives or codes of practice: organisational guidelines and procedures; recognised compliance agency/body's standards, directives or codes of practice; equipment manufacturer's operating specification/range; customer standards and requirements; national and/or International standards or directives; health, safety and environmental Requirements)	1	3	-	-
PC20. deal promptly and effectively with problems within your control and seek help and guidance from the relevant people if you have problems that you cannot resolve	2	2	-	-
PC21. ensure that the designs are protected in line with organizational procedures	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Evaluate engineering design options</i>	6	12	-	-
PC22. obtain clear criteria on which to base the evaluation (Criteria for evaluating designs: function; financial constraints; manufacturing or installation requirements; installation or commissioning requirements; building redundancy into the design; appropriate materials; technology; aesthetics; performance/capability; reliability; life cycle of product, system or process; compatibility; maintenance and repair; product features; availability of resources; characteristics; corporate branding; components to be used; any interface requirements; future customer support; timescales; diversity/alternatives; safety; environmental/sustainability factors)	1	3	-	-
PC23. obtain the necessary information from the available sources	1	3	-	-
PC24. evaluate the design against the established criteria, using appropriate evaluation methods (Evaluation methods: market research; software simulation; analysis of the design documentation; simulation; model; prototype assessment; pilot trial; small-scale production)	2	3	-	-
PC25. make recommendations on various design options, and communicate the results of the evaluation to the relevant people	2	3	-	-
NOS Total	35	65	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0407
NOS Name	Create and evaluate mechanical engineering design options
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Design
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	24/11/2017
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Qualification Pack

CSC/N0408: Make or modify 3D mechanical engineering models using CAD system

Description

This unit is about creating or modifying 3D mechanical engineering models using CAD system. The candidate will be able to extract all necessary information in order to carry out the modelling operations based on model brief or a request for a change/modification; produce 3D CAM code files, managing files, tools and installed software.

Scope

This unit/task covers the following:

- Prepare for 3D mechanical engineering modelling using CAD system
- Create and make changes to 3D mechanical engineering models using CAD system

Elements and Performance Criteria

Prepare for 3D mechanical engineering modelling using CAD system

To be competent, the user/individual on the job must be able to:

- PC1.** plan the modelling activities before starting them
- PC2.** use appropriate sources to obtain the required information (Required information: model brief/request, specifications, change order/modification request, regulations, manuals, sample component, calculations, previous models/designs, sketches, notes from meetings/discussions, standards reference documents (such as limits and fits, tapping drill charts), other available data)
- PC3.** access and use the correct modelling software and tools (Modelling software and tools: solid modelling, wire frame modelling, surface modelling)
- PC4.** check that all the equipment is correctly connected and in a safe and usable working condition
- PC5.** power up the equipment and activate the appropriate modelling tools
- PC6.** set up the modelling environment and select a suitable template/folder
- PC7.** set up and check that all peripheral devices are connected and correctly operating (such as keyboard, mouse, light pen, digitizer/tablet, scanner, printer, plotter)
- PC8.** set the drawing datum at a convenient point to create a modelling template with title, file number, material, date
- PC9.** establish coordinate system, orientation and views as per the job requirement

Create and make changes to 3D mechanical engineering models using CAD system

To be competent, the user/individual on the job must be able to:

- PC10.** create entities in 3D space as per job requirement
- PC11.** modify entities in 3D space as per job requirement
- PC12.** create 3-D views on the screen by manipulating drawing planes and inserting 3-D geometric shapes
- PC13.** creating swept, extruded and revolved solids in 3-D space

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- PC14.** produce sectioned models (cutting planes and cross hatching)
- PC15.** use pre-drawn library files and primitives to produce a 3-D model
- PC16.** extracting mass and area properties from solid model
- PC17.** identify and use key features of solid modelling software package to produce models (Key features: extrude, extrude cut, solid model, mirror, revolve, wire frame, radius/chamfer, hide, rib, rectangular pattern, fillet, cut/remove, circular pattern, shell, development view, motion analysis, animation, defining material property, exploded views)
- PC18.** perform drawing for solid modelling
- PC19.** extract physical properties as per job requirement, including volume, mass and centre of gravity
- PC20.** take into account the following factors, as appropriate to the model being produced (Factors: function, cost, physical space, quality, lifetime of the product, operating environment, manufacturing method, tolerances, interfaces, ergonomics, clearance, safety, materials, aesthetics, apply rendering techniques)
- PC21.** use pan, isometric and zoom CAD operations to highlight design areas in the modelling environment
- PC22.** modify parts in the assembly environment using the following features (Features: constrained parts and assemblies, straight lines, insertion of standard components, hidden detail, dimensions, symbols and abbreviations, hatching and shading, angular surfaces, curved surfaces, parts lists, text, circles or ellipses, material color, surface texture)
- PC23.** produce 3-D drawings incorporating section views with all necessary annotation
- PC24.** produce a model for export to the following manufacturing systems (Manufacturing systems: DNC (Direct Numerically controlled) /CNC (Computer Numerically controlled) machines; 3D printer; other specific system)
- PC25.** produce models which comply with organizational guidelines; statutory regulations and codes of practice; CAD software standards; national and international standards
- PC26.** confirm that the model is as per job specifications and contains all relevant information
- PC27.** use appropriate techniques to create models that are sufficiently and clearly detailed
- PC28.** use codes and other references that follow the required conventions
- PC29.** make sure that models are checked and approved by the appropriate person
- PC30.** save the models in the appropriate file type and location
- PC31.** produce hard copies of the finished models, with sufficient detail to allow production
- PC32.** deal promptly and effectively with problems within your control, and seek help and guidance from the relevant people if you have problems that you cannot resolve
- PC33.** shut down the CAD system to a safe condition on completion of the modelling activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** specific health, safety and environmental requirements that apply to the product or process to be designed
- KU2.** the limits of their own authority, and to whom they should report if they have problems that they cannot resolve

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- KU3.** importance of establishing and recording responsibilities
- KU4.** organizational procedures and information systems for storing drawing data
- KU5.** relevant sources and methods for obtaining any required technical information relevant to the model being produced (such as drawing briefs, specification sheets, request for changes or modifications to models; technical information such as limits and fits, contraction allowances, bearing selection, surface finish)
- KU6.** identification of the correct 3D drawing software package from the menu or windows environment; the various techniques that are available to access and use the CAD software (such as mouse, menu or tool bar, light pens, digitizers and tablets, printers or plotters, and scanners)
- KU7.** how to access the specific computer modelling software to be used, and the use of the help file to aid efficient operation of the relevant drawing system
- KU8.** documentation required for particular applications (such as design briefs, specification sheets, request for change orders)
- KU9.** types of drawings that may be produced by the modelling software
- KU10.** how to set up the viewing screen to show multiple views of the component to help with drawing creation (to include isometric front and side elevations)
- KU11.** national, international and organizational standards and conventions that are used for the models/drawings
- KU12.** application and use of modelling tools (such as for straight lines, curves and circles; how to add dimensions and text to drawings)
- KU13.** how to access, recognize and use a wide range of standard components and symbol libraries from the CAD equipment
- KU14.** applications of different 3D modelling programs such as surface modelling, solid modelling, wire frame modelling
- KU15.** how to produce models with sufficient information to allow them to be successfully exported to the manufacturing system used
- KU16.** need for document control (such as ensuring that completed models are approved, labelled and stored on a suitable storage medium)
- KU17.** need to create backup copies, and to file them in a separate and safe location, also filing and storing hard copies for use in production
- KU18.** how to produce hard copies of the drawings, and the advantages and disadvantages of printers and plotters
- KU19.** purpose for which the 3D model is to be developed
- KU20.** appropriate coordinate system for the job
- KU21.** reasons for selecting the chosen coordinate system
- KU22.** orientation of the model with respect to the coordinate system
- KU23.** number of views required to establish the model
- KU24.** procedures for creating entities in 3D space
- KU25.** entities that can be created/manipulated in 3D space
- KU26.** procedures for manipulating entities in 3D space
- KU27.** procedures for creating ruled and revolved surfaces in 3D space
- KU28.** applications of ruled and revolved surfaces
- KU29.** procedures for modifying existing 3D models

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- KU30.** procedures for saving drawing files
- KU31.** various formats in which drawing files can be saved
- KU32.** reasons for using different formats when saving drawing files
- KU33.** procedures for extracting data with respect to the physical properties of shapes created in 3D space
- KU34.** physical properties of shapes created in 3D space that can be extracted from the drawing file
- KU35.** erecting of exploded views
- KU36.** creating own toolbox
- KU37.** dynamic simulation of models creating intelligent models using parametric modelling
- KU38.** producing composite models (composite regions and composite solids)
- KU39.** producing sectioned models (cutting planes and cross hatching)
- KU40.** using pre-drawn library files and primitives to produce a 3-D model
- KU41.** extracting mass and area properties from solid model
- KU42.** applying rendering techniques to a 3D model (rendering types and preferences, render lighting techniques, and views and scenes)
- KU43.** using various materials and surface finish options
- KU44.** producing hard copies of 3-D solid models
- KU45.** saving 3-D models in various file formats for retrieval into other CAD application software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
- GS3.** undertake numerical operations, and calculations/ formulae (Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
- GS4.** identify and draw various basic, compound and solid shapes as per dimensions given (Basic shapes: square, rectangle, triangle, circle) (Compound shapes: involving squares, rectangles, triangles, circles, semi-circles, quadrants of a circle) (Solid shapes: cube, rectangular prism, cylinder)
- GS5.** use appropriate units and number systems to express degree of accuracy (Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity)
- GS6.** interpret and express tolerance in terms of limits on dimensions
- GS7.** calculation of the value of angles in a triangle (Angles in a triangle: right-angled, isosceles, equilateral)
- GS8.** convey and share technical information clearly using appropriate language
- GS9.** check and clarify task-related information
- GS10.** liaise with appropriate authorities using correct protocol

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- GS11.** communicate with people in respectful form and manner in line with organizational protocol
- GS12.** listen to questions and concerns of the customer and provide resolution in a respectful manner as per organizational guidelines
- GS13.** use basic office applications like spread sheet, word processor, presentations
- GS14.** use ERP software and other organizational software specific to quality function
- GS15.** use email to communicate within the organization as per organization guidelines
- GS16.** be well dressed and groomed
- GS17.** put forward ones point of view in a convincing manner
- GS18.** plan, prioritize and sequence work operations as per job requirements
- GS19.** organize and analyze information relevant to work
- GS20.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS21.** communicate with customers following organizational protocols and practices generating customer satisfaction and delight
- GS22.** undertake clear and open communication with customers for trust building and clarifying and managing expectations of customers
- GS23.** respond to customer expectation promptly and recognizing and communicating limits of ones authority
- GS24.** deal with customer feedback
- GS25.** handle customer disgruntlement and dissatisfaction
- GS26.** work in a team in order to achieve better results
- GS27.** identify and clarify work roles within a team
- GS28.** communicate and cooperate with others in the team for better results
- GS29.** seek assistance from fellow team members
- GS30.** co-ordinate across teams and personnel for getting work done
- GS31.** work taking responsibility for own work outcomes
- GS32.** adhere to work timings, dress code and other organizational policies
- GS33.** work following laid down rules, procedures, instructions and policies
- GS34.** conduct oneself express dissent during conflict situations while exercising restraint
- GS35.** avoid and manage distractions to be disciplined at work
- GS36.** work by time management for achieving better results
- GS37.** identify problems with work planning, procedures, output and behavior and their implications
- GS38.** prioritize and plan for problem solving
- GS39.** communicate problems appropriately to others
- GS40.** identify sources of information and support for problem solving
- GS41.** seek assistance and support from other sources to solve problems
- GS42.** identify effective resolution techniques
- GS43.** select and apply resolution techniques
- GS44.** seek evidence for problem resolution
- GS45.** inspect quality of own or other employee's work
- GS46.** analyze information according to enterprise and work requirements

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- GS47.** use diagnostic skills to identify and determine causes of faults, including interpretation of in-built fault indicators and error codes
- GS48.** take decisions within if within own jurisdiction or take approval for case outside own jurisdiction
- GS49.** work towards achieving better results for self, others and organization
- GS50.** display initiative and enterprise
- GS51.** undertake and express new ideas and initiatives to others
- GS52.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS53.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS54.** achieve more by applying ones competencies in new and different situations and contexts to achieve more
- GS55.** identify potential business opportunities for the company
- GS56.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS57.** participate in on-the-job and other learning, training and development interventions and assessments
- GS58.** clarify task related information with appropriate personnel or technical adviser
- GS59.** seek to improve and modify own work practices

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for 3D mechanical engineering modelling using CAD system</i>	8	21	-	-
PC1. plan the modelling activities before starting them	1	2	-	-
PC2. use appropriate sources to obtain the required information (Required information: model brief/request, specifications, change order/modification request, regulations, manuals, sample component, calculations, previous models/designs, sketches, notes from meetings/discussions, standards reference documents (such as limits and fits, tapping drill charts), other available data)	1	2	-	-
PC3. access and use the correct modelling software and tools (Modelling software and tools: solid modelling, wire frame modelling, surface modelling)	1	2	-	-
PC4. check that all the equipment is correctly connected and in a safe and usable working condition	-	2	-	-
PC5. power up the equipment and activate the appropriate modelling tools	1	1	-	-
PC6. set up the modelling environment and select a suitable template/folder	1	3	-	-
PC7. set up and check that all peripheral devices are connected and correctly operating (such as keyboard, mouse, light pen, digitizer/tablet, scanner, printer, plotter)	1	3	-	-
PC8. set the drawing datum at a convenient point to create a modelling template with title, file number, material, date	1	3	-	-
PC9. establish coordinate system, orientation and views as per the job requirement	1	3	-	-
<i>Create and make changes to 3D mechanical engineering models using CAD system</i>	20	51	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. create entities in 3D space as per job requirement	1	2	-	-
PC11. modify entities in 3D space as per job requirement	1	1	-	-
PC12. create 3-D views on the screen by manipulating drawing planes and inserting 3-D geometric shapes	1	3	-	-
PC13. creating swept, extruded and revolved solids in 3-D space	1	3	-	-
PC14. produce sectioned models (cutting planes and cross hatching)	1	3	-	-
PC15. use pre-drawn library files and primitives to produce a 3-D model	-	2	-	-
PC16. extracting mass and area properties from solid model	1	2	-	-
PC17. identify and use key features of solid modelling software package to produce models (Key features: extrude, extrude cut, solid model, mirror, revolve, wire frame, radius/chamfer, hide, rib, rectangular pattern, fillet, cut/remove, circular pattern, shell, development view, motion analysis, animation, defining material property, exploded views)	-	2	-	-
PC18. perform drawing for solid modelling	1	2	-	-
PC19. extract physical properties as per job requirement, including volume, mass and centre of gravity	1	2	-	-
PC20. take into account the following factors, as appropriate to the model being produced (Factors: function, cost, physical space, quality, lifetime of the product, operating environment, manufacturing method, tolerances, interfaces, ergonomics, clearance, safety, materials, aesthetics, apply rendering techniques)	1	2	-	-
PC21. use pan, isometric and zoom CAD operations to highlight design areas in the modelling environment	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. modify parts in the assembly environment using the following features (Features: constrained parts and assemblies, straight lines, insertion of standard components, hidden detail, dimensions, symbols and abbreviations, hatching and shading, angular surfaces, curved surfaces, parts lists, text, circles or ellipses, material color, surface texture)	1	2	-	-
PC23. produce 3-D drawings incorporating section views with all necessary annotation	1	2	-	-
PC24. produce a model for export to the following manufacturing systems (Manufacturing systems: DNC (Direct Numerically controlled) /CNC (Computer Numerically controlled) machines; 3D printer; other specific system)	1	2	-	-
PC25. produce models which comply with organizational guidelines; statutory regulations and codes of practice; CAD software standards; national and international standards	1	2	-	-
PC26. confirm that the model is as per job specifications and contains all relevant information	1	2	-	-
PC27. use appropriate techniques to create models that are sufficiently and clearly detailed	1	2	-	-
PC28. use codes and other references that follow the required conventions	1	2	-	-
PC29. make sure that models are checked and approved by the appropriate person	-	2	-	-
PC30. save the models in the appropriate file type and location	-	2	-	-
PC31. produce hard copies of the finished models, with sufficient detail to allow production	1	2	-	-
PC32. deal promptly and effectively with problems within your control, and seek help and guidance from the relevant people if you have problems that you cannot resolve	1	3	-	-
PC33. shut down the CAD system to a safe condition on completion of the modelling activities	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	28	72	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0408
NOS Name	Make or modify 3D mechanical engineering models using CAD system
Sector	Capital Goods
Sub-Sector	Machine Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery
Occupation	Design
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	24/11/2017
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Qualification Pack

CSC/N1335: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment.

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuffless (without folds), trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors Equipment: hand shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace (Hazards: sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces(sharp, slippery, uneven, chipped, broken, etc.); hazardous substances(chemicals, gas, oxy-fuel, fumes, dust, etc.); physical hazards(working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc.) electrical hazards (power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.) Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness))
- PC5.** carry out safe working practices while dealing with hazards to ensure the safety of self and others (Safe working practices: using protective clothing and equipment; putting up and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter, spillage and unsafe object lying casually; while working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc.; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working in confined places, trenches or at heights, etc. including safety harness, fall arrestors, etc.)
- PC6.** state methods of accident prevention in the work environment of the job role (Methods of accident prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors)
- PC7.** state location of general health and safety equipment in the workplace (General health and safety equipment: fire extinguishers; first aid equipment; safety instruments and clothing; safety installations(eg fire exits, exhaust fans))

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- PC8.** inspect for faults, set up and safely use steps and ladders in general use (Ladder faults: corrosion of metal components, deterioration, splits and cracks timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.) (Ladders set up: firm/level base, clip/lash down, leaning at the correct angle, etc.)
- PC9.** work safely in and around trenches, elevated places and confined areas
- PC10.** lift heavy objects safely using correct procedures
- PC11.** apply good housekeeping practices at all times (Good housekeeping practices: clean/tidy work areas, removal/disposal of waste products, protect surfaces)
- PC12.** identify common hazard signs displayed in various areas (Various areas: on chemical containers; equipment; packages; inside buildings; in open areas and public spaces, etc.)
- PC13.** retrieve and/or point out documents that refer to health and safety in the workplace (Documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (eg government notices))

Fire safety

To be competent, the user/individual on the job must be able to:

- PC14.** use the various appropriate fire extinguishers on different types of fires correctly (Types of fires: Class A: eg. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: eg. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity); Class D: combustible metals such as magnesium, titanium, and sodium (These fires burn at extremely high temperatures and require special suppression agents))
- PC15.** demonstrate rescue techniques applied during fire hazard
- PC16.** demonstrate good housekeeping in order to prevent fire hazards
- PC17.** demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC18.** demonstrate how to free a person from electrocution
- PC19.** administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC20.** demonstrate basic techniques of bandaging
- PC21.** respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments
- PC22.** perform and organize loss minimization or rescue activity during an accident in real or simulated environments
- PC23.** administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC24.** demonstrate the artificial respiration and the CPR Process
- PC25.** participate in emergency procedures (Emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work)
- PC26.** complete a written accident/incident report or dictate a report to another person, and send report to person responsible (Incident Report includes details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified)

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Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC27. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
- KU2.** names and location of documents that refer to health and safety in the workplace
- KU3.** meaning of hazards and risks
- KU4.** health and safety hazards commonly present in the work environment and related precautions
- KU5.** possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU6.** possible causes of risk and accident(Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness))
- KU7.** methods of accident prevention(Methods of accident prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors)
- KU8.** safe working practices when working with tools and machines
- KU9.** safe working practices while working at various hazardous sites
- KU10.** where to find all the general health and safety equipment in the workplace
- KU11.** various dangers associated with the use of electrical equipment
- KU12.** preventative and remedial actions to be taken in the case of exposure to toxic materials(Exposure: ingested, contact with skin, inhaled); (Preventative action: ventilation, masks, protective clothing/ equipment); (Remedial action: immediate first aid, report to supervisor Toxic materials: solvents, flux, lead)
- KU13.** importance of using protective clothing/equipment while working
- KU14.** precautionary activities to prevent the fire accident
- KU15.** various causes of fire(Causes of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.)
- KU16.** techniques of using the different fire extinguishers
- KU17.** different methods of extinguishing fire
- KU18.** different materials used for extinguishing fire(Materials: sand, water, foam, CO2, dry powder)
- KU19.** rescue techniques applied during a fire hazard
- KU20.** various types of safety signs and what they mean
- KU21.** appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU22.** content of written accident report
- KU23.** potential injuries and ill health associated with incorrect manual handling

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KU24. safe lifting and carrying practices

KU25. personal safety, health and dignity issues relating to the movement of a person by others

KU26. potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and comprehend basic content to read labels, charts, signages

GS2. read and comprehend basic English to read manuals of operations

GS3. read an accident/incident report in local language or English

GS4. write an accident/incident report in local language or English

GS5. question co-workers appropriately in order to clarify instructions and other issues

GS6. give clear instructions to coworkers, subordinates others

GS7. make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines

GS8. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity

GS9. remain congenial while discussing and debating issues with co-workers

GS10. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice

GS11. ask for, provide and receive required assistance where possible to ensure achievement of work related objectives

GS12. thank co-workers for any assistance received

GS13. offer appropriate respect based on mutuality and respect for fellow workmanship and authority

GS14. identify immediate or temporary solutions to resolve delays

GS15. identify sources of support that can be availed of for problem solving for various kind of problems

GS16. seek appropriate assistance from other sources to resolve problems

GS17. report problems that you cannot resolve to appropriate authority

GS18. identify cause and effect relations in their area of work

GS19. use cause and effect relations to anticipate potential problems and their solution

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	21	32	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuffless (without folds), trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors Equipment: hand shields, machine guards, residual current devices, shields, dust sheets, respirator)	1	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	1	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	1	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace (Hazards: sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces(sharp, slippery, uneven, chipped, broken, etc.); hazardous substances(chemicals, gas, oxy-fuel, fumes, dust, etc.); physical hazards(working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc.) electrical hazards (power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.) Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness))	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. carry out safe working practices while dealing with hazards to ensure the safety of self and others (Safe working practices: using protective clothing and equipment; putting up and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter, spillage and unsafe object lying casually; while working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc.; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working in confined places, trenches or at heights, etc. including safety harness, fall arrestors, etc.)	2	2	-	-
PC6. state methods of accident prevention in the work environment of the job role (Methods of accident prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors)	2	1	-	-
PC7. state location of general health and safety equipment in the workplace (General health and safety equipment: fire extinguishers; first aid equipment; safety instruments and clothing; safety installations(eg fire exits, exhaust fans))	2	3	-	-
PC8. inspect for faults, set up and safely use steps and ladders in general use (Ladder faults: corrosion of metal components, deterioration, splits and cracks timber components, imbalance, loose rungs, missing/unfixed nuts or bolts, etc.) (Ladders set up: firm/level base, clip/lash down, leaning at the correct angle, etc.)	2	3	-	-
PC9. work safely in and around trenches, elevated places and confined areas	2	3	-	-
PC10. lift heavy objects safely using correct procedures	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. apply good housekeeping practices at all times (Good housekeeping practices: clean/tidy work areas, removal/disposal of waste products, protect surfaces)	2	3	-	-
PC12. identify common hazard signs displayed in various areas (Various areas: on chemical containers; equipment; packages; inside buildings; in open areas and public spaces, etc.)	1	2	-	-
PC13. retrieve and/or point out documents that refer to health and safety in the workplace (Documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (eg government notices))	1	3	-	-
<i>Fire safety</i>	4	10	-	-
PC14. use the various appropriate fire extinguishers on different types of fires correctly (Types of fires: Class A: eg. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: eg. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity); Class D: combustible metals such as magnesium, titanium, and sodium (These fires burn at extremely high temperatures and require special suppression agents))	1	2	-	-
PC15. demonstrate rescue techniques applied during fire hazard	1	2	-	-
PC16. demonstrate good housekeeping in order to prevent fire hazards	1	3	-	-
PC17. demonstrate the correct use of a fire extinguisher	1	3	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	20	-	-
PC18. demonstrate how to free a person from electrocution	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC20. demonstrate basic techniques of bandaging	1	2	-	-
PC21. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC22. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC23. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC24. demonstrate the artificial respiration and the CPR Process	1	2	-	-
PC25. participate in emergency procedures (Emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work)	1	3	-	-
PC26. complete a written accident/incident report or dictate a report to another person, and send report to person responsible (Incident Report includes details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified)	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	2	2	-	-
PC27. demonstrate correct method to move injured people and others during an emergency	2	2	-	-
NOS Total	36	64	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1335
NOS Name	Use basic health and safety practices at the workplace
Sector	Capital Goods
Sub-Sector	Machine Tools, Process Plant Machinery, Dies, Moulds and Press Tools, Electrical and Power Machinery, Plastics Manufacturing Machinery, Light Engineering Goods, Textile Manufacturing Machinery
Occupation	Machining
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CSC/N1336: Work effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening etc.

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working (Communication etiquette: do not use abusive language; use appropriate titles and terms of respect; do not eat or chew while talking (vice versa) etc.)
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace (Disciplined behaviors: e.g. punctuality; completing tasks as per given time and standards; not gossiping and idling time; eliminating waste, honesty, etc.)
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and coordinate within the organization
- KU6.** importance of effective communication in the workplace

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- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication
- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read basic terms and terminologies to accurately interpret work-related documents, labels, supervisor instructions in the local language
- GS2.** read and interpret accurate information from various relevant work instructions and records
- GS3.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS4.** write basic numbers, quantities and work-related terminology for operational requirements in the local language
- GS5.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans, and other associated requirements
- GS6.** give clear instructions to co-workers about the type of output required and answer queries
- GS7.** display active listening skills while interacting with co-workers and others in the workplace
- GS8.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS9.** take steps within ones limits of authority to initiate modification in plan if the circumstances require it
- GS10.** check that work meets customer requirements
- GS11.** deliver consistent and reliable service to internal and external customers
- GS12.** work with co-workers and supervisor to resolve any issues that threaten disruption, increase risk, cause delays or under-achievement of quality and targets as per the planned schedule

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working (Communication etiquette: do not use abusive language; use appropriate titles and terms of respect; do not eat or chew while talking (vice versa) etc.)	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace (Disciplined behaviors: e.g. punctuality; completing tasks as per given time and standards; not gossiping and idling time; eliminating waste, honesty, etc.)	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1336
NOS Name	Work effectively with others
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Machining
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	30/09/2021
Next Review Date	30/09/2024
NSQC Clearance Date	30/09/2021

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for the Qualification Pack will be created by CGSC.
2. Performance Criteria (PC) have been assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
3. The assessment for the theory part will/may be based on knowledge bank of questions approved CGSC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. Assessment Agencies will create Assessor Guides comprising of Theory and Practical Assessment Set and Guidelines for each examination/training centre (as per assessment criteria below). The same will be approved by CGSC for adequacy.
6. To successfully attain Certification on the Qualification Pack, the trainee must score a minimum of 70% in each Core NOS and minimum of 50% in all non-core NOS. In addition, a candidate needs to attain a minimum overall pass percentage of 70% for certification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CSC/N0402.Make or modify 2D mechanical engineering drawings using CAD system	30	70	-	-	100	20
CSC/N0405.Identify customers requirement and create an engineering design brief	36	64	-	-	100	12
CSC/N0406.Develop plan for engineering design process	38	62	-	-	100	12
CSC/N0407.Create and evaluate mechanical engineering design options	35	65	-	-	100	12
CSC/N0408.Make or modify 3D mechanical engineering models using CAD system	28	72	-	-	100	24
CSC/N1335.Use basic health and safety practices at the workplace	36	64	-	-	100	10
CSC/N1336.Work effectively with others	30	70	-	-	100	10
Total	233	467	-	-	700	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CNC	Computer Numerically Controlled
CAD	Computer Aided Design
2D	2 Dimensional
3D	3 Dimensional
CO₂	Carbon Dioxide
CPR	Cardiac Pulmonary Resuscitation
ISO	International Organization For Standardization
PPE	Personal Protective Equipment
CD	Compact Disc
DVD	Digital Video Disc Or Digital Versatile Disc

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.